

ECE 647: Lecture 16

First-order Sufficient Condition:

- Suppose that the set $C \subset E$ is convex and that the function $f : C \rightarrow R$ is convex.

Let $\bar{x} \in C$. If the condition $f'(\bar{x}; x - \bar{x}) \geq 0$ holds for all $x \in C$, then $\bar{x}$ is a global minimizer of f on C .

- If in addition f is differentiable, then the sufficient condition is

$$-\nabla f(\bar{x}) \in N_C(\bar{x})$$

Same as the necessary condition!

Note that in general, this statement does not hold for non-convex problems.

Proof

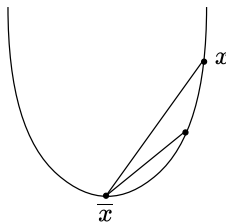
Again, note that

$$\frac{f(\bar{x} + t(x - \bar{x})) - f(\bar{x})}{t}$$

is decreasing as $t \downarrow 0$. The limit as $t \downarrow 0$ is $f'(\bar{x}; x - \bar{x})$. Hence,

$$f(\bar{x} + t(x - \bar{x})) - f(\bar{x}) \geq t \cdot f'(\bar{x}; x - \bar{x}) \geq 0$$

Let $t = 1$, we have $f(\bar{x}) \leq f(x)$. The result then follows.



Examples

① Unconstrained Convex Problems:

- $C = R^N$, f is convex
- Necessary and sufficient conditions for differentiable functions:

$$\nabla f(\bar{x}) = 0$$

Note: If f is not differentiable, use subgradients.

② Single Convex Constraint:

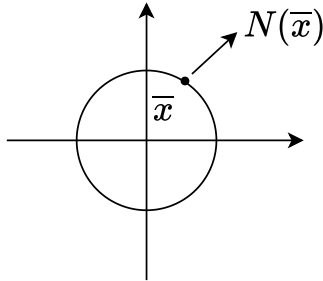
$$\begin{aligned} &\min f_0(x) \\ &\text{sub to } x^T x \leq 1 \end{aligned}$$

- either at the interior

$$\nabla f_0(\bar{x}) = 0 \text{ \& } \bar{x}^T \bar{x} < 1$$

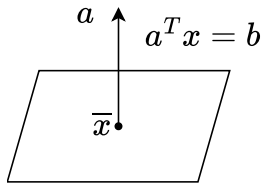
- or at the boundary $\bar{x}^T \bar{x} = 1$
 - The gradient of $\bar{x}^T \bar{x}$ is $\bar{x}$.
 - Hence

$$\nabla f_0(\bar{x}) = \lambda \bar{x}, \lambda \geq 0 \text{ \& } \bar{x}^T \bar{x} = 1$$



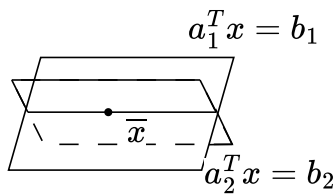
③ Equality-Constrained Convex Problems:

- $C = \{x \mid Ax = b\}$
- If b is a scalar (i.e., only 1 constraint $a^T x = b$), then C is a hyper-plane.
The normal vector is $A^T = a$ or $-A^T = -a$

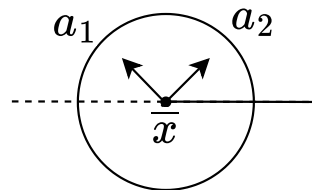


$$\Rightarrow -\nabla f_0(\bar{x}) = \lambda a \quad \lambda \in \mathbb{R}$$

- If b is a vector, the normal cone will be bigger



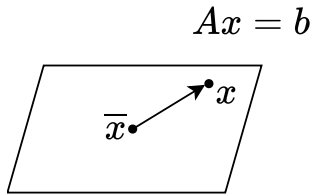
(a)



(b)

- Spans all linear combinations of a_1 & a_2

- In general, the normal cone contains all vectors of the form $A^T v$
 - To see this, consider any $x \in C$
 - $\Rightarrow Ax = b, A\bar{x} = b$
 - $\Rightarrow A(x - \bar{x}) = 0$
 - Such vectors $x - \bar{x}$ are said to be in the null space of A
 - If $d = A^T v$
 - Then $d^T(x - \bar{x}) = v^T A(x - \bar{x}) = 0$



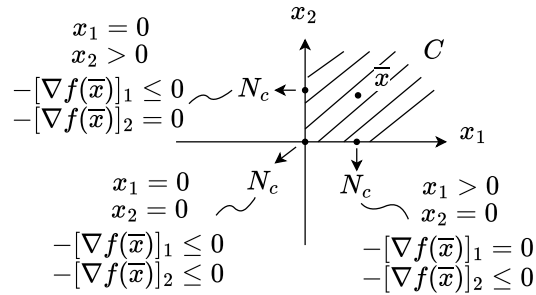
- We thus have

$$\begin{cases} \nabla f(\bar{x}) + A^T v = 0 \\ A\bar{x} = b \end{cases}$$

- More on this with Lagrange duality.

④ Minimization over the non-negative gradients

$$\begin{aligned} & \min f_0(x) \\ & \text{sub to } x \geq 0 \end{aligned}$$



- We observe that
 $-\nabla f(\bar{x}) \geq 0, \bar{x} \geq 0$
 -Further

$$\nabla f(\bar{x}_i) \cdot \bar{x}_i = 0$$

for each i

- This is called complementary slackness conditions

The sets of indices that correspond to non-zero elements of the vector $\bar{x}$ and $\nabla f(\bar{x})$ have empty intersections.

In general, $\bar{x}$ is local minimizer

$$\begin{aligned}
 &\Leftrightarrow \nabla f(\bar{x})^T (y - \bar{x}) \geq 0 \quad \forall y \geq 0 \\
 &\Leftrightarrow \nabla f(\bar{x}) \geq 0, -\nabla f(\bar{x})^T \cdot \bar{x} \geq 0, \bar{x} \geq 0 \\
 &\quad \quad \quad \uparrow \quad \quad \quad \uparrow \\
 &\quad \quad \quad y \rightarrow +\infty \quad y = 0 \\
 &\Leftrightarrow \nabla f(\bar{x})_i \cdot \bar{x}_i = 0 \quad \forall i \\
 &\nabla f(\bar{x}) \geq 0 \\
 &\bar{x} \geq 0
 \end{aligned}$$